

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

**MECHANICAL
PROPERTIES OF
FLUIDS**

LECTURE-05 PHYSICS

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Topics To Be Covered

- 1 Revision
- 2 Terminal Velocity
- 3 Surface Tension
- 4 Some Phenomenon which are Explain on the Basic of Surface Tension
- 5 ~~Some Phenomenon which are Explain on the Basic of Surface Tension~~

RECAP FROM 1ST LECTURE

Fluid

liquid

gas

Density ρ

$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Atmospheric pressure P_0

Variation of pressure with depth



$$P_B = P_A + \rho gh$$

Pascal Law

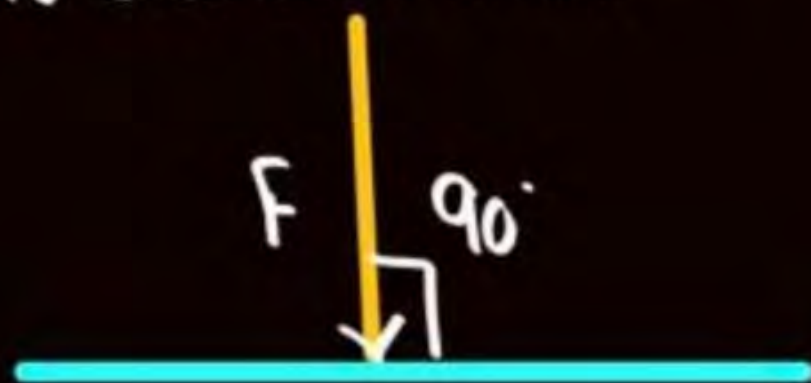
Recap-2

① Application pascal law

- Hydraulic Brake ✓
- Hydraulic lift

② Thrust

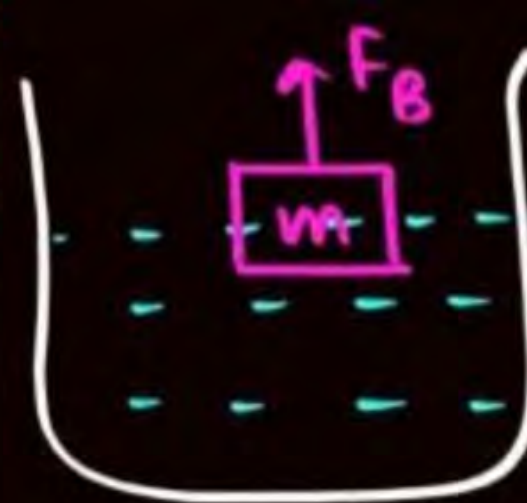
- Any perpendicular force is called thrust



Buoyant force



Archimedes principle



$$F_B = \rho V g$$

Flow

Streamline flow

$$Re < 2000$$

turbulent flow

$$Re \gg \gg 2000$$

Equation of Continuity

$$a_1 v_1 = a_2 v_2$$

Recep - ③

Bernoullies Theorm

- law of conservation of Energy

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

Application of B-Theorm

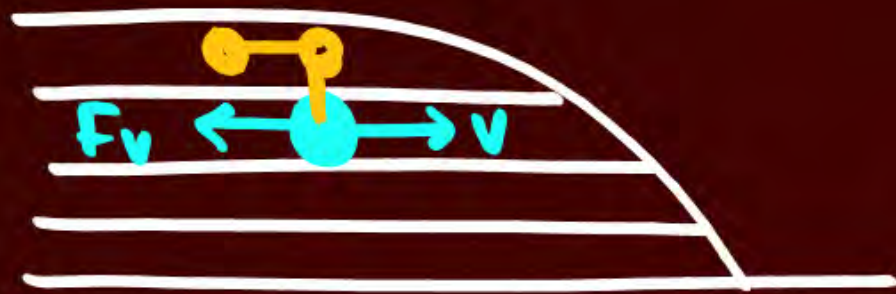
① Spray / Atomizer

② Magnus effect

③ lift of aeroplane

RECAP FROM 4TH LECTURE

• Viscosity



$$F_v = -\eta A \frac{dv}{dx}$$

$\eta \Rightarrow$ coefficient of viscosity

η = Pascal sec (SI)

η = poise (c.g.s)

Dimensional formula of η

$$\checkmark \eta = m l^{-1} t^{-1}$$

Stoke law

$$F_v = 6\pi\eta r v$$

$$a=1$$

$$b=1$$

$$c=1$$

Terminal Velocity

lecture . no . 5



Atmosphere



Gas

Fluid

$u=0$

$g = -9.8 \text{ m/s}^2$

$a = -9.8 \text{ m/s}^2$

$s = 1000 \text{ m} = 1 \text{ km}$

$v^2 = u^2 + 2as$ ①

$v^2 = 0^2 + 2 \times -9.8 \times 1000 \text{ m}$

$v^2 = -19.6 \times 1000$

$v^2 = -19.6 \times 10 \times 100$

$v^2 = -196 \times 100$

$v = \sqrt{-196 \times 100}$

$v = \sqrt{-14^2 \times 10^4}$

$v = -14 \times 10$

$v = -140 \text{ m/s}$

$v = ? - 140 \text{ m/s}$
 504 km/hr

Shazia (Takli)



②

$$V = -140 \text{ m/s}$$

$$V = -140 \times 1 \text{ m/s} \quad \underline{1 \text{ m/s}} = \underline{3.6 \text{ km/hr}}$$

$$V = -140 \times 3.6 \text{ km/hr}$$

$$V = \underline{-504 \text{ km/hr}} \quad \phi$$

$$V = \underline{10 \text{ km/hr}} - \underline{5 \text{ km/hr}}$$

Terminal Velocity (2026) ✓ 2024

- The final constant velocity with which the spherical Body (Ball) moves in Fluid (gas & liquid)

Gas

Fluid

$$F_{net} = 0$$

$$W = F_b + F_v$$

$$u = 0$$

$$F_b \uparrow \quad F_v = 6\pi\eta r v \uparrow \uparrow$$

$$v \uparrow \uparrow \uparrow$$

$$W \uparrow \uparrow \uparrow \uparrow$$

$$F_b \uparrow \quad F_v = 6\pi\eta r v_T$$

$$A$$

$$v_T = ?$$



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Derivation

• At point (A) $\{F_{net} = 0\}$

$$W = F_b + F_v \quad W = mg \quad F_b = \rho V g$$

$$\textcircled{\times} \textcircled{mg} = \rho V g + 6\pi\eta r v_T \quad F_v = 6\pi\eta r \textcircled{v_T}?$$

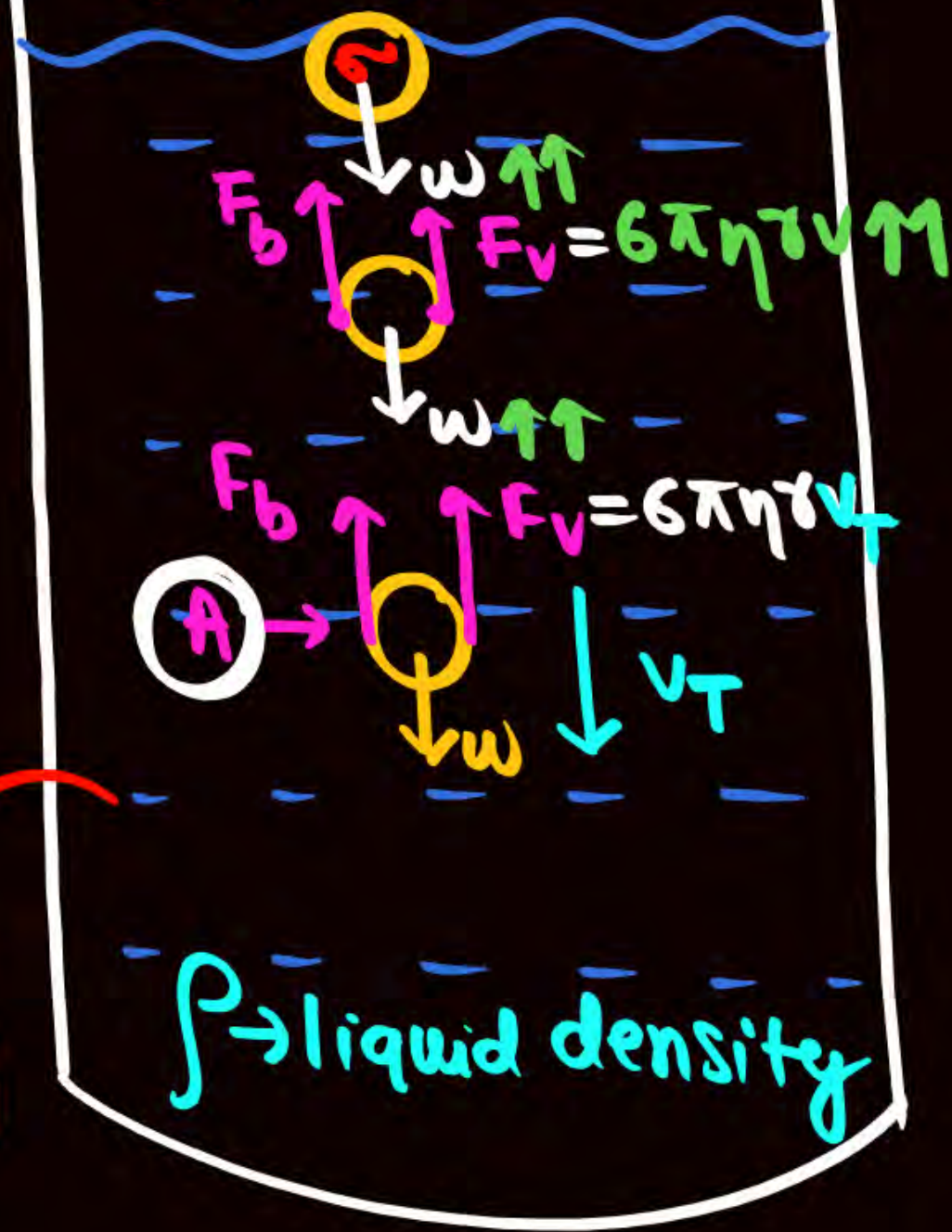
$$\rho V g = \rho V g = 6\pi\eta r v_T$$

$$\rho V g - \rho V g = 6\pi\eta r v_T$$

$$v g (\rho - \rho) = 6\pi\eta r v_T$$

density
of Body

$$\rho = \frac{m}{V} \quad (4)$$
$$\rho V = m$$



liquid
(Fluid)

Derivation

$$\underline{V}g(\sigma - \rho) = 6\pi\eta r v_T$$

V  Volume of sphere = $\frac{4}{3}\pi r^3$

$$\frac{4}{3}\pi r^3 g(\sigma - \rho) = \underline{6\pi\eta r v_T}$$

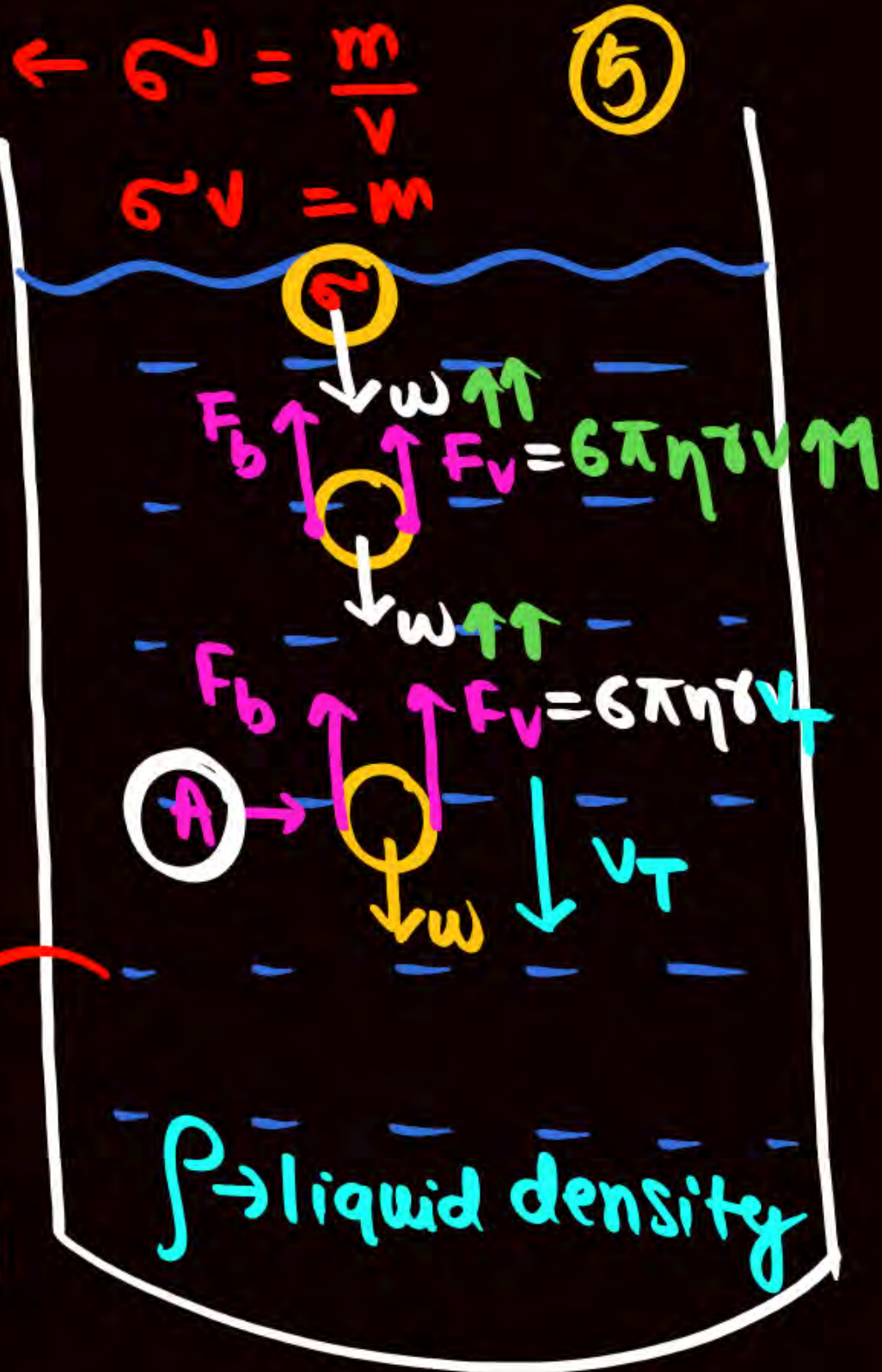
$$\frac{2}{3}\pi r^2 g(\sigma - \rho) = v_T$$

$$\frac{2r^2 g(\sigma - \rho)}{9\eta} = v_T$$

density
of Body

$$\leftarrow \sigma = \frac{m}{V} \quad (5)$$
$$\sigma V = m$$

liquid
(Fluid)



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$$V_T = \frac{2r^2g(\sigma - p)}{q\eta}$$

$$V_T = \frac{2}{q} \frac{r^2g(\sigma - p)}{\eta}$$

$$F_{\text{net}} = 0$$

$$ma = 0$$

$$a = \frac{0}{m} = \underbrace{0 \text{ m/s}^2}$$

$$a = \frac{v_f - v_i}{t}$$

$$0 \times t = v_f - v_i$$

$$0 = v_f - v_i$$

$$\underline{v_i = v_f}$$

Surface Tension

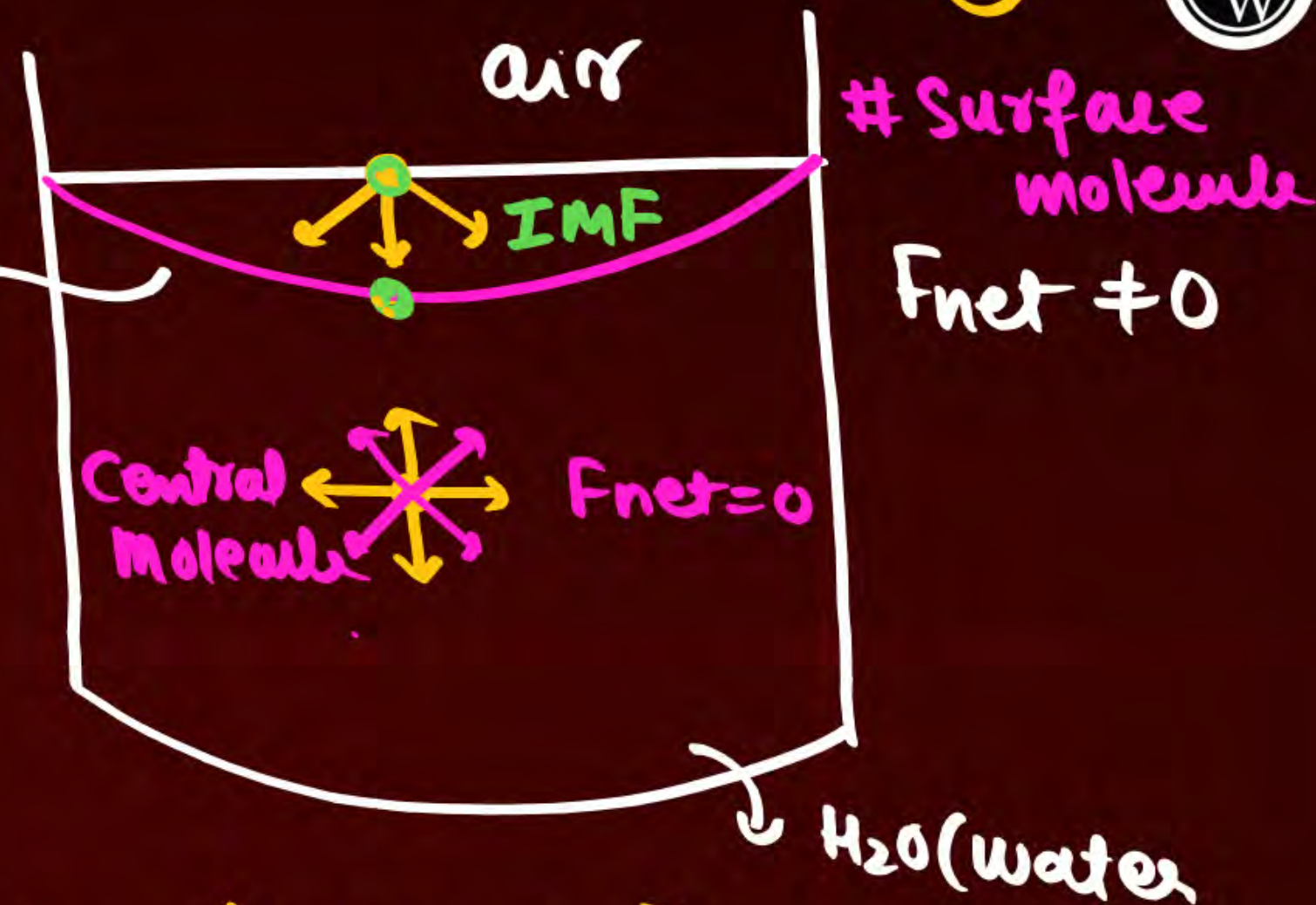
ST / T

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⇒ The unbalanced Force acting per unit length on the liquid surface is Max Surface tension

Concave



Surface tension $\propto$ Intere molecular force

$$\tau / ST / \text{Surface tension} = \frac{F}{l}$$

SI unit = $\frac{N}{m}$



Q Find out the Dimensional formula of $ST = ?$

⑧

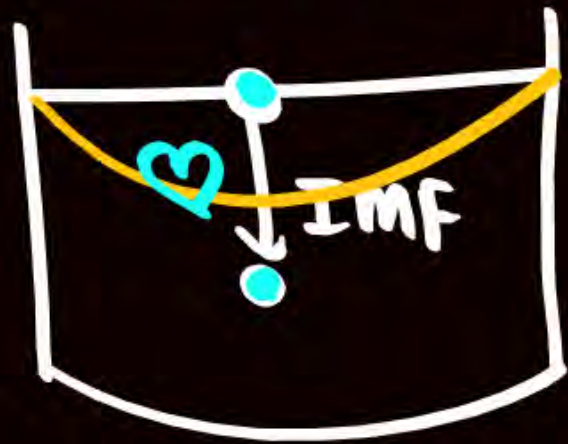
So $\cancel{T}/ST = \frac{F}{\cancel{q}}$

$$\cancel{T}/ST = \frac{MLT^{-2}}{A} = \underline{M^1 T^{-2}}$$

Factors that effect the Surface tension

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① Intermolecular force (2023)



The Intermolecular force is directly Proportional to Surface tension
 $\uparrow \text{IMF} \propto \text{ST}$

② Temperature



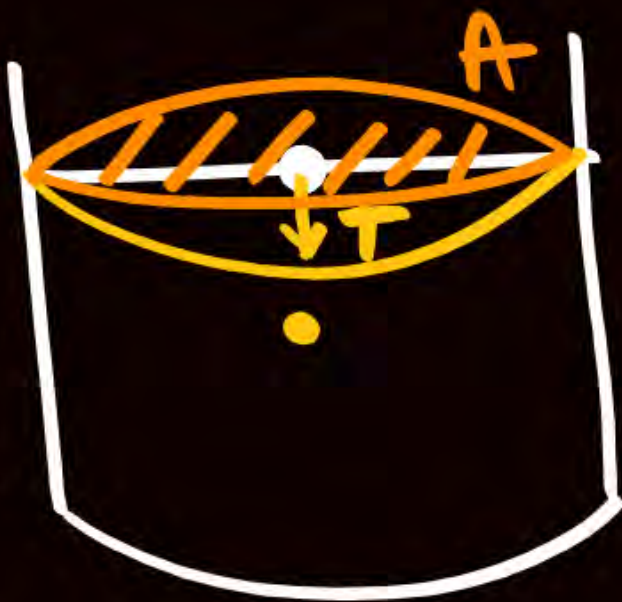
$\uparrow \uparrow \uparrow$ heat

Temperature is inversely proportional to IMF and Surface tension

Temp $\uparrow \uparrow$ IMF $\downarrow \downarrow$ ST $\downarrow \downarrow$

Surface Energy (SE)

- The potential energy store on liquid surface due to surface tension. This is Plan Surface Energy.



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$$\underline{SE = TA}$$

$$SE \propto T \text{ --- (I)}$$

$$SE \propto A \text{ --- (II)}$$

$$SE \propto TA \quad K \rightarrow \text{Energy Constant}$$

$$SE = KTA \quad K \rightarrow 1$$

$$SE = 1 \times TA = \underline{TA}$$

Neet 2021

Dimensional formula of Surface energy?

$$ML^2T^{-2}$$

Sol

$$EE = TA \quad A = L^2$$

$$= MT^{-2}L^2$$

$$= \underline{ML^2T^{-2}}$$

SOME Important phenomenon
due to Surface Tension



HOMework



Derive $v_T = ?$
(5mark)

2021

SUMMARY

Surface Tension
Terminal Velocity

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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